

Project 1: Linear System

File Structure

File	Role
Vector.h / Vector.cpp	1D vector class with full operator support
Matrix.h / Matrix.cpp	2D matrix class with determinant, inverse, pseudo-inverse
LinearSystem.h / LinearSystem.cpp	Gaussian elimination + Conjugate Gradient solver
LinearRegression.h / .cpp	Part B: dataset loader, least-squares regression, RMSE/R2
main.cpp	Test driver — runs all unit tests then Part B pipeline
Makefile	Build system (Windows-compatible via MSYS2)

Project 2: Smart E-Commerce System with Recommendation Engine

File Structure

File	Role
Product.h / Product.cpp	Product entity with snapshot, stock management, and serialization features.
Cart.h / Cart.cpp	Shopping cart containing CartItem struct, supporting add/remove/update operations.
Order.h / Order.cpp	Completed order record and serialization management.
User.h / User.cpp	Abstract base class designed for the authentication system.
Customer.h / Customer.cpp	Inherits from User; includes Cart, OrderHistory, and interaction tracking mechanisms.
Admin.h / Admin.cpp	Inherits from User; handles product management and statistical reporting.

RecommendationEngine.h / .cpp	Scoring algorithm processing: $\text{views} * 0.2 + \text{purchases} * 0.8 + \text{categoryBonus}$
FileManager.h / FileManager.cpp	Centralized file I/O operations for products, users, orders, and user interactions.
ECommerceSystem.h / .cpp	Main controller managing menus, authentication, and orchestrating the entire system workflow.
main.cpp	System entry point.
Makefile	Build system configuration.